Ashutosh Rana

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AI engineer reading two concurrent degrees, Smart Manufacturing Engineering at Example Institute of Technology and Data Science and AI at Example University. I build machine learning that ends up attached to a sensor or a process rather than to a notebook, and every result I report comes with the baseline it beat and the conditions under which it fails.

Open to AI/ML internships for Summer 2027.

EXPERIENCE

Undergraduate Research Assistant, Applied ML Lab, Example University
Research / India / On-site / 08.2025 - now

- Instrumented three CNC spindles with accelerometers and built the acquisition path from Arduino to a Postgres time-series table.
- Trained a bearing-fault classifier on 41 hours of labelled vibration data. Baseline was a fixed RMS threshold at 0.71 F1; the model reached 0.89 F1 on a held-out machine it had never seen.
- Wrote the failure-mode note the lab now hands to new students: the model degrades below 900 RPM because the fault signature falls under the noise floor.

Competition Contributor, Kaggle
Part-time / Remote / 01.2025 - now

- Took part in three competitions, owning data cleaning and preprocessing for a two-person team.
- Merged five source datasets into one feature table and wrote a row-for-row equivalence test against the previous pipeline before switching over.
- Best finish: 539 of 3,300 entrants. The gap to the top 100 was feature engineering, not model choice, which is written up in the notebook.

Machine Learning Intern, Nuvora Analytics
Internship / Ahmedabad, India / Remote / 05.2026 - 07.2026

- Built the review-triage classifier that now routes support tickets. 12,000 labelled reviews, 0.91 macro F1 against a 0.68 keyword baseline.
- Quantized the deployed model to INT8, holding accuracy inside 0.4 points and dropping p95 latency from 240ms to 88ms.
- Shipped the eval harness alongside it, so the next person can tell whether a change helped.

Volunteer Teacher, Example Literacy Program
Volunteer / India / On-site / 06.2023 - 04.2024

- Taught basic language proficiency to domestic helpers and homemakers, weekly, across one academic year.
- Twelve learners. Nine could read a utility bill unaided by the end, which was the only outcome we tracked.

SELECTED SYSTEMS

Spindle Sentinel
Deployed / 09.2025 - now

- Bearing-fault detection on live CNC spindles, running in the lab since January and catching two failures before they stopped a machine.
- 0.89 macro F1 on a held-out machine, 41h labelled vibration data, 3 fault classes, against 0.71 with a fixed RMS threshold.
- Python / scikit-learn / Arduino / PostgreSQL / Spectrograms

Grounded
Shipped / 02.2026 - now

- Retrieval over four semesters of lecture notes that answers with citations, or says it does not know.
- 0.84 faithfulness on 120 hand-labelled QA pairs, against 0.52 with embedding-only retrieval. Refusal rate on unanswerable questions went from 0 to 88 percent.
- Python / BM25 / Embeddings / RRF / FastAPI

Review Triage
Deployed / 05.2026 - 07.2026

- Sentiment and intent classifier routing incoming support reviews, replacing a keyword rule set.
- 0.91 macro F1 on 12,000 labelled reviews across 5 classes, against 0.68 for the rules it replaced. INT8 serving cut p95 latency from 240ms to 88ms for 0.4 points of accuracy.
- Python / Hugging Face / Fine-tuning / INT8 / Docker

Edge Ladder Prototype / 03.2026 - now

- A measured comparison of four precisions on one model, so the accuracy-for-latency trade stops being a guess.
- INT8 gives 2.7x throughput for 0.4 points of accuracy. INT4 gives 3.9x for 6.1 points, which no product should accept.
- PyTorch / Quantization / Benchmarking

EDUCATION

B.Tech, Computer Science, Example Institute of Technology 2024 - 2028

- Manufacturing systems, control and instrumentation. Taken concurrently with the degree below.

BS (Hons), Data Science and AI, Example University 2024 - 2028

- Statistics, ML theory, and the mathematics under the libraries.

Example Secondary School Class XII 2024 / Class X 2022

SKILLS AND TOOLS

Languages

- Python, Java, C++, C, SQL

ML and data

- scikit-learn, NumPy, Pandas, Matplotlib, exploratory data analysis

Serving and infra

- PostgreSQL, Hugging Face, Docker, Bash, Git

Applied AI

- Prompt engineering, retrieval, eval harnesses, quantization

Hardware

- Arduino, sensor data acquisition, robotics, signal processing

Foundations

- Linear algebra, probability, statistics, optimization

CERTIFICATIONS

Kaggle 2025

- Python / Intro to Machine Learning / Pandas